

Database System Engineering and Distributed Backend Development

WEEK-3

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Step 1 — Switch to the BookFlow Database

```
use bookflow
```

The screenshot shows the MongoDB Atlas Data Explorer interface. The top navigation bar indicates the organization is 'GUNAPATI HANSIKA...' and the project is 'Project 0'. The left sidebar shows the 'Data Explorer' view with options for 'My Queries', 'Data Modeling', and 'CLUSTERS (1)'. The main panel displays the 'Cluster0' database with a table listing collections. The 'bookflow' collection is highlighted.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
book_metadata	-	4.10 kB	0 B	0	0 B	1	4.10 kB

Step 2 — Create the book_metadata Collection

```
db.createCollection("book_metadata")
```

The screenshot shows the MongoDB Atlas Data Explorer interface. The top navigation bar indicates the organization is 'GUNAPATI HANSIKA...' and the project is 'Project 0'. The left sidebar shows the 'Data Explorer' view with options for 'My Queries', 'Data Modeling', and 'CLUSTERS (1)'. The main panel displays the 'Cluster0' database with a table listing collections. The 'book_metadata' collection is highlighted.

Cluster0 > bookflow > book_metadata

Documents (0) Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

It only

Step 3 — Insert Document 1 (Embedded + Referenced Pattern)

```
db.book_metadata.insertOne({
  book_id: 201,
  title: "The Silent Patient",
  author: "Alex Michaelides",
  published_year: 2022,
  category: "Thriller",
  format: "Digital EPUB",
  file_size: "2.8 MB",
  member_id: 101,
  reviews: [
    { member_id: 101, reviewer_name: "Hansika", rating: 5, comments: "Excellent psychological thriller" },
    { member_id: 104, reviewer_name: "Aditya Rao", rating: 4, comments: "Very gripping with an unexpected twist" },
    { member_id: 105, reviewer_name: "Sneha Kapoor", rating: 5, comments: "Excellent narration and pacing" }
  ]
})
```

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'Data Explorer' sidebar is visible, showing the 'book_metadata' collection under 'Cluster0'. The main panel displays the 'Documents' tab for the 'book_metadata' collection. A single document is shown in a table format:

ObjectID	book_id	title	author	published_year	category	format
1	201	"The Silent Patient"	"Alex Michaelides"	2022	"Thriller"	"Digital EPUB"

Below the table, there are buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE'. A tooltip 'Export query to language' is visible over the 'EXPORT CODE' button. The interface also includes a search bar, a query editor, and various navigation and action buttons.

Step 4 — Insert Document 2 (Digital Format)

```
db.book_metadata.insertOne({
  book_id: 202,
  title: "Atomic Habits",
  author: "James Clear",
  published_year: 2023,
  category: "Self Help",
  format: "Digital PDF",
  file_size: "3.4 MB",
  member_id: 102,
  reviews: [
```

```

    { member_id: 102, reviewer_name: "Hasini", rating: 5, comments: "Excellent practical habit building guide" },
    { member_id: 106, reviewer_name: "Riya Sharma", rating: 4, comments: "Good but a little repetitive later on" },
    { member_id: 101, reviewer_name: "Hansika", rating: 5, comments: "Excellent life changing concepts" }
  ]
})

```

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar shows the 'bookflow' database and the 'book_metadata' collection selected. The main panel displays the 'Documents' tab for 'Cluster0 > bookflow > book_metadata'. It shows two documents in a JSON format. The first document has a book_id of 201, title 'The Silent Patient', author 'Alex Michaelides', published_year 2022, category 'Thriller', format 'Digital EPUB', file_size '2.8 MB', member_id 101, and reviews. The second document has a book_id of 202, title 'Atomic Habits', author 'James Clear', published_year 2023, category 'Self Help', format 'Digital PDF', file_size '3.4 MB', member_id 102, and reviews.

Step 5 — Insert Document 3 (Physical Format / Flexible Schema)

```

db.book_metadata.insertOne({
  book_id: 203,
  title: "Wings of Fire",
  author: "A. P. J. Abdul Kalam",
  published_year: 2019,
  category: "Autobiography",
  format: "Physical",
  dimensions: { height: "22 cm", width: "14 cm", weight: "350 g" },
  member_id: 103,
  reviews: [
    { member_id: 103, reviewer_name: "Bindhu", rating: 5, comments: "Excellent inspiring autobiography" },
    { member_id: 107, reviewer_name: "Kunal Mehta", rating: 4, comments: "Very motivational read for students" }
  ]
})

```

ORGANIZATION GUNAPATI HANSIKA... PROJECT Project 0

Cluster0 book_metadata bookflow +

Data Explorer

{ My Queries

Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

```
{
  "book_id": 201,
  "title": "The Silent Patient",
  "author": "Alex Michaelides",
  "published_year": 2022,
  "category": "Thriller",
  "format": "Digital EPUB",
  "file_size": "2.8 MB",
  "member_id": 101,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 202,
  "title": "Atomic Habits",
  "author": "James Clear",
  "published_year": 2023,
  "category": "Self Help",
  "format": "Digital PDF",
  "file_size": "3.4 MB",
  "member_id": 102,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 203,
  "title": "Wings of Fire",
  "author": "A. P. J. Abdul Kalam",
  "published_year": 2019,
  "category": "Autobiography",
  "format": "Physical",
  "dimensions": {},
  "member_id": 103,
  "reviews": []
}
```

Step 6 — Verify Inserted Documents

```
db.book_metadata.find().pretty()
```

ORGANIZATION GUNAPATI HANSIKA... PROJECT Project 0

Cluster0 book_metadata bookflow +

Data Explorer

{ My Queries

Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3 Aggregations Schema Indexes 1 Validation

{ }

ADD DATA BULK EXPORT CODE

```
{
  "_id": {},
  "book_id": 201,
  "title": "The Silent Patient",
  "author": "Alex Michaelides",
  "published_year": 2022,
  "category": "Thriller",
  "format": "Digital EPUB",
  "file_size": "2.8 MB",
  "member_id": 101,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 202,
  "title": "Atomic Habits",
  "author": "James Clear",
  "published_year": 2023,
  "category": "Self Help",
  "format": "Digital PDF",
  "file_size": "3.4 MB",
  "member_id": 102,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 203,
  "title": "Wings of Fire",
  "author": "A. P. J. Abdul Kalam",
  "published_year": 2019,
  "category": "Autobiography",
  "format": "Physical",
  "dimensions": {},
  "member_id": 103,
  "reviews": []
}
```

Step 7 — Comparison Query: \$gt (Rating Greater Than 4)

```
db.book_metadata.find({
  "reviews.rating": { $gt: 4 }
})
```

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'Data Explorer' sidebar displays the 'CLUSTERS (1)' section with a search bar and a tree view containing 'Cluster0' and its sub-clusters: 'admin', 'bankdb', 'bookflow', 'book_metadata', 'local', and 'sample_mflix'. The 'book_metadata' cluster is selected. The main panel shows the 'Cluster0 > bookflow > book_metadata' view. The 'Documents' tab is active, showing a query filter: `{ "reviews.rating": { "$gt": 4 } }`. Below the filter are buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE'. The results pane displays three document entries, each with a play button icon. The first document is for 'The Silent Patient' (book_id: 201, member_id: 101), the second is for 'Atomic Habits' (book_id: 202, member_id: 102), and the third is for 'Wings of Fire' (book_id: 203, member_id: 103).

```
{
  "_id": {},
  "book_id": 201,
  "title": "The Silent Patient",
  "author": "Alex Michaelides",
  "published_year": 2022,
  "category": "Thriller",
  "format": "Digital EPUB",
  "file_size": "2.8 MB",
  "member_id": 101,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 202,
  "title": "Atomic Habits",
  "author": "James Clear",
  "published_year": 2023,
  "category": "Self Help",
  "format": "Digital PDF",
  "file_size": "3.4 MB",
  "member_id": 102,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 203,
  "title": "Wings of Fire",
  "author": "A. P. J. Abdul Kalam",
  "published_year": 2019,
  "category": "Autobiography",
  "format": "Physical",
  "dimensions": {},
  "member_id": 103,
  "reviews": []
}
```

Step 8 — \$in Query (Rating 4 or 5)

```
db.book_metadata.find({
  "reviews.rating": { $in: [4, 5] }
})
```

ORGANIZATION
GUNAPATI HANSIKA...

PROJECT
Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

Cluster0

book_metadata

bookflow

Cluster0 > bookflow > book_metadata

Documents 3

Aggregations

Schema

Indexes 1

Validation

{ "reviews.rating": { "\$in": [4, 5] } }

+ ADD DATA

BULK

EXPORT CODE

{

"_id": {},

"book_id": 201,

"title": "The Silent Patient",

"author": "Alex Michaelides",

"published_year": 2022,

"category": "Thriller",

"format": "Digital EPUB",

"file_size": "2.8 MB",

"member_id": 101,

"reviews": [{}]

}

{

"_id": {},

"book_id": 202,

"title": "Atomic Habits",

"author": "James Clear",

"published_year": 2023,

"category": "Self Help",

"format": "Digital PDF",

"file_size": "3.4 MB",

"member_id": 102,

"reviews": [{}]

}

{

"_id": {},

"book_id": 203,

"title": "Wings of Fire",

"author": "A. P. J. Abdul Kalam",

"published_year": 2019,

"category": "Autobiography",

"format": "Physical",

"dimensions": {},

"member_id": 103,

}

Step 9 — Regex Query (Digital Format)

```
db.book_metadata.find({
  format: { $regex: "Digital", $options: "i" }
})
```

ORGANIZATION

GUNAPATI HANSIKA...

PROJECT

Project 0

Cluster0

book_metadata

bookflow

+

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3

Aggregations

Schema

Indexes 1

Validation

{ "format": { "\$regex": "Digital", "\$options": "i" } }

ADD DATA

BULK

EXPORT CODE

{

"_id": { ... },

"book_id": 201,

"title": "The Silent Patient",

"author": "Alex Michaelides",

"published_year": 2022,

"category": "Thriller",

"format": "Digital EPUB",

"file_size": "2.8 MB",

"member_id": 101,

"reviews": [...]

}

▶ {

"_id": { ... },

"book_id": 202,

"title": "Atomic Habits",

"author": "James Clear",

"published_year": 2023,

"category": "Self Help",

"format": "Digital PDF",

"file_size": "3.4 MB",

"member_id": 102,

"reviews": [...]

}

Step 10 — Aggregation Pipeline: Top Rated Books Report

```

db.book_metadata.aggregate([
  { $match: { published_year: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  { $group: {
    _id: "$title",
    average_rating: { $avg: "$reviews.rating" },
    total_reviews: { $sum: 1 }
  } },
  { $sort: { average_rating: -1 } }
])

```

ORGANIZATION GUNAPATI HANSIKA... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3 Aggregations Schema Indexes 1 Validation

\$match \$unwind \$group \$sort Edit

ALL RESULTS EXPORT CODE

```

_id: "The Silent Patient"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Atomic Habits"
average_rating: 4.666666666666667
total_reviews: 3

```

Step 11 — Create a Text Index on Review Comments

```
db.book_metadata.createIndex({
  "reviews.comments": "text"
})
```

ORGANIZATION GUNAPATI HANSIKA... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3 Aggregations Schema Indexes 2 Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
id	REGULAR	36.9 kB	5 (since Sat Aug 22 2026)	UNIQUE	READY
reviews.comments_text	TEXT	20.5 kB	0 (since Sat Aug 22 2026)		READY

Step 12 — Text Search Query

```
db.book_metadata.find({
  $text: { $search: "excellent" }
})
```


ORGANIZATION GUNAPATI HANSIKA... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 3 Aggregations Schema Indexes 2 Validation

```
{ "$text": { "$search": "excellent" } }
```

ADD DATA BULK EXPORT CODE

```
{
  "_id": {},
  "book_id": 201,
  "title": "The Silent Patient",
  "author": "Alex Michaelides",
  "published_year": 2022,
  "category": "Thriller",
  "format": "Digital EPUB",
  "file_size": "2.8 MB",
  "member_id": 101,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 202,
  "title": "Atomic Habits",
  "author": "James Clear",
  "published_year": 2023,
  "category": "Self Help",
  "format": "Digital PDF",
  "file_size": "3.4 MB",
  "member_id": 102,
  "reviews": []
}
```

```
{
  "_id": {},
  "book_id": 203,
  "title": "Wings of Fire",
  "author": "A. P. J. Abdul Kalam",
  "published_year": 2019,
  "category": "Autobiography",
  "format": "Physical",
  "dimensions": {},
  "member_id": 103,
  "reviews": []
}
```

Step 13 — explain() Query Performance Analysis

```
db.book_metadata.find({
  $text: { $search: "excellent" }
}).explain("executionStats")
```

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Interpret

Visual Tree Raw Output



Query Performance Summary

- 3 documents returned
- 3 documents examined
- 0 ms execution time
- Is not sorted in memory
- 3 index keys examined

Query used the following index:

`_fts (text)`
`_ftsx`

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree Raw Output

```
{
  "explainVersion": "1",
  "queryPlanner": {
    "namespace": "bookflow.book_metadata",
    "parsedQuery": {
      "$text": {
        "$search": "excellent",
        "$language": "english",
        "$caseSensitive": false,
        "$diacriticSensitive": false
      }
    },
    "indexFilterSet": false,
    "queryHash": "CF6F4CEE",
    "planCacheShapeHash": "CF6F4CEE",
    "planCacheKey": "08852285",
    "optimizationTimeMillis": 0,
    "maxIndexedOrSolutionsReached": false,
    "maxIndexedAndSolutionsReached": false,
    "maxScansToExplodeReached": false,
    "prunedSimilarIndexes": false,
    "winningPlan": {
      "isCached": false,
      "stage": "TEXT_MATCH",
      "indexPrefix": {},
      "indexName": "reviews.comments_text",
      "parsedTextQuery": {
        "terms": ["excel"],
        "negatedTerms": [],
        "phrases": [],
        "negatedPhrases": []
      },
      "textIndexVersion": 3,
      "inputStage": {
        "stage": "FETCH",
        "inputStage": {
          "stage": "IXSCAN",
          "keyPattern": {
```